



# 12316 - Probing the UV to MIR spectra of local Little Red Dots

Cycle: 5, Proposal Category: GO

## INVESTIGATORS

| <i>Name</i>                        | <i>Institution</i>                                |
|------------------------------------|---------------------------------------------------|
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## OBSERVATIONS

| <i>Folder</i>      | <i>Observation</i> | <i>Label</i>           | <i>Observing Template</i>           | <i>Science Target</i> |
|--------------------|--------------------|------------------------|-------------------------------------|-----------------------|
| Observation Folder |                    |                        |                                     |                       |
|                    | 1                  | J1022 NIRSPEC IFU      | NIRSpec IFU Spectroscopy            | (1) SDSSJ1022+0841    |
|                    | 2                  | J1025 NIRSPEC IFU      | NIRSpec IFU Spectroscopy            | (2) SDSSJ1025+1402    |
|                    | 3                  | J1047 NIRSPEC IFU      | NIRSpec IFU Spectroscopy            | (3) SDSSJ1047+0739    |
|                    | 4                  | J1340 NIRSPEC IFU      | NIRSpec IFU Spectroscopy            | (4) SDSSJ1340+0211    |
|                    | 5                  | J1717 NIRSPEC IFU      | NIRSpec IFU Spectroscopy            | (5) SDSSJ1717+3807    |
|                    | 6                  | TOL2240 NIRSPEC IFU    | NIRSpec IFU Spectroscopy            | (6) TOL2240-384       |
|                    | 7                  | J1022 MIRI MRS         | MIRI Medium Resolution Spectroscopy | (1) SDSSJ1022+0841    |
|                    | 8                  | J1025 MIRI MRS         | MIRI Medium Resolution Spectroscopy | (2) SDSSJ1025+1402    |
|                    | 9                  | J1047 MIRI MRS         | MIRI Medium Resolution Spectroscopy | (3) SDSSJ1047+0739    |
|                    | 10                 | J1340 MIRI MRS         | MIRI Medium Resolution Spectroscopy | (4) SDSSJ1340+0211    |
|                    | 11                 | J1717 MIRI MRS         | MIRI Medium Resolution Spectroscopy | (5) SDSSJ1717+3807    |
|                    | 12                 | Tol2240 MIRI MRS       | MIRI Medium Resolution Spectroscopy | (6) TOL2240-384       |
|                    | 13                 | Tol2240 MIRI MRS B KGD | MIRI Medium Resolution Spectroscopy | (6) TOL2240-384       |

## **ABSTRACT**

JWST has uncovered a mysterious population of compact, red sources at  $z \sim 2-10$  known as "Little Red Dots" (LRDs). Their observed properties defy nearly all typical AGN properties, leading researchers to hypothesize that they could be primordial supermassive stars or enshrouded supermassive black holes ('quasi-stars'), offering a direct window into heavy black hole seed formation. A small population of local analogs at  $z < 0.5$  have been recently identified,  $\sim 5-10$  magnitudes brighter than those from JWST, providing an unparalleled window into the phenomena. We propose JWST/NIRSpec+MIRI + HST UV spectroscopy of six local LRD analogs to decipher their nature. Rest-frame 0.1-0.2 and 1–28  $\mu\text{m}$  spectra will probe key diagnostics: UV emission, absorption and outflows; dust continuum shape; silicate features; PAH emission; and fine-structure lines, and numerous absorption lines and bands. This sample will push further into the MIR than is possible for their distant cousins, thereby enabling novel insights into the enigmatic LRD population and their central power sources.

## **OBSERVING DESCRIPTION**

We propose to observe six local LRDs ( $z \sim 0.07-0.2$ ) with NIRSpec IFU (G140H/F100LP, G235H/F170LP, G395H/F290LP) and MIRI IFU high-resolution spectroscopy (MRS, all setups). For the scene, assume the dominant nuclear component is unresolved. In the case of J1047, we model a faint cigar-like extended component based on previous HST imaging, while for Tol2240 and J1340, we do not due to limited knowledge from ground-based imaging. We expect similar ( $\sim 5\%$ ) faint narrow components around all objects; these should not hinder background assessment. We assign the actual UV-MIR SED to the dominant nuclear component, while for J1047 we adopt a M33 (Wolf-Rayet galaxy) SED template for the extended component, normalized in r-band to the extended/total light ratio. We seek  $S/N \sim 50$  across much of the contiguous 1-5.1  $\mu\text{m}$  coverage, as assessed for the a 0.3" radius aperture centered on the nucleus. For MIRI MRS, we aim to achieve  $SNR \sim 30$  in all but the final low-sensitivity setup ( $SNR \sim 10$ ), which will allow assessment of faint absorption lines and bands.

We adopt a 4-point dither, with NRSIRS2RAPID or SLOWR1 readout, allowing us to achieve background subtraction within the target observations themselves. As the targets are point-like with good positions, we do not opt for target acquisition imaging; guide star precision is adequate. We adopt 1 integration and tune the groups to achieve the SNR desired.

# Proposal 12316 - Targets - Probing the UV to MIR spectra of local Little Red Dots

| Fixed Targets | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Name           | Target Coordinates                                                                   | Targ. Coord. Corrections                | Miscellaneous |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------|-----------------------------------------|---------------|
|               | (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SDSSJ1022+0841 | RA: 10 22 8.5286 (155.5355358d)<br>Dec: +08 41 56.19 (8.69894d)<br>Equinox: J2000    | Parallax: 0"<br>Epoch of Position: 2016 |               |
|               | Comments: This object was generated by the targetselector and retrieved from the NED database.<br>Extent is <1"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES                                                                                                                                                                                                                                                                                                                                                               |                |                                                                                      |                                         |               |
|               | (2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SDSSJ1025+1402 | RA: 10 25 30.2919 (156.3762162d)<br>Dec: +14 02 7.40 (14.03539d)<br>Equinox: J2000   | Parallax: 0"<br>Epoch of Position: 2016 |               |
|               | Comments: This object was generated by the targetselector and retrieved from the NED database.<br>Extent is ~0.5"x1"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES                                                                                                                                                                                                                                                                                                                                                          |                |                                                                                      |                                         |               |
|               | (3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SDSSJ1047+0739 | RA: 10 47 55.9266 (161.9830275d)<br>Dec: +07 39 51.23 (7.66423d)<br>Equinox: J2000   | Epoch of Position: 2016                 |               |
|               | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.<br>Extragalactic objec at z=0.168. PM not relevant. Extent is ~0.5"x1.0"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES |                |                                                                                      |                                         |               |
|               | (4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SDSSJ1340+0211 | RA: 13 40 52.0334 (205.2168058d)<br>Dec: +02 11 57.27 (2.19924d)<br>Equinox: J2000   | Epoch of Position: 2016                 |               |
|               | Comments: This object was generated by the targetselector and retrieved from the NED database.<br>Extent is ~1-2"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES                                                                                                                                                                                                                                                                                                                                                             |                |                                                                                      |                                         |               |
|               | (5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SDSSJ1717+3807 | RA: 17 17 41.7363 (259.4239013d)<br>Dec: +38 07 52.47 (38.13124d)<br>Equinox: J2000  | Epoch of Position: 2016                 |               |
|               | Comments: This object was generated by the targetselector and retrieved from the NED database.<br>Extent is <1"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES                                                                                                                                                                                                                                                                                                                                                               |                |                                                                                      |                                         |               |
|               | (6)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | TOL2240-384    | RA: 22 43 32.9629 (340.8873454d)<br>Dec: -38 11 20.74 (-38.18909d)<br>Equinox: J2000 | Epoch of Position: 2016                 |               |
|               | Comments: Extent is 1"x2.5". Pointing center shifted NE by ~0.3" relative to peak, so IFU will cover point source and NE extension (based on LS DR10 imaging)<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES                                                                                                                                                                                                                                                                                                                 |                |                                                                                      |                                         |               |

# Proposal 12316 - Observation 1 - Probing the UV to MIR spectra of local Little Red Dots

|                   |                                                                                             |                |                                                                                                |            |                  |         |                          |         |               |                    |                     |                 |                              |
|-------------------|---------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------|------------|------------------|---------|--------------------------|---------|---------------|--------------------|---------------------|-----------------|------------------------------|
| Observation       | Proposal 12316, Observation 1: J1022 NIRSPEC IFU                                            |                |                                                                                                |            |                  |         |                          |         |               |                    |                     |                 | Tue May 26 16:00:29 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                  |                |                                                                                                |            |                  |         |                          |         |               |                    |                     |                 |                              |
| Diagnostics       | Observing Template: NIRSpec IFU Spectroscopy                                                |                |                                                                                                |            |                  |         |                          |         |               |                    |                     |                 |                              |
|                   | (Visit 1:1) Warning (Form): Data Excess over lower threshold                                |                |                                                                                                |            |                  |         |                          |         |               |                    |                     |                 |                              |
| Fixed Targets     | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                |                                                                                                |            |                  |         |                          |         |               |                    |                     |                 |                              |
|                   | #                                                                                           | Name           | Target Coordinates                                                                             |            |                  |         | Targ. Coord. Corrections |         |               |                    | Miscellaneous       |                 |                              |
| Template          | (1)                                                                                         | SDSSJ1022+0841 | RA: 10 22 8.5286 (155.5355358d)                                                                |            |                  |         | Parallax: 0"             |         |               |                    |                     |                 |                              |
|                   |                                                                                             |                | Dec: +08 41 56.19 (8.69894d)                                                                   |            |                  |         | Epoch of Position: 2016  |         |               |                    |                     |                 |                              |
| Dithers           |                                                                                             |                | Equinox: J2000                                                                                 |            |                  |         |                          |         |               |                    |                     |                 |                              |
|                   |                                                                                             |                | Comments: This object was generated by the targetselector and retrieved from the NED database. |            |                  |         |                          |         |               |                    |                     |                 |                              |
| Spectral Elements |                                                                                             |                | Extent is <1"                                                                                  |            |                  |         |                          |         |               |                    |                     |                 |                              |
|                   |                                                                                             |                | Category=Galaxy                                                                                |            |                  |         |                          |         |               |                    |                     |                 |                              |
| Dithers           |                                                                                             |                | Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]          |            |                  |         |                          |         |               |                    |                     |                 |                              |
|                   |                                                                                             |                | Extended=YES                                                                                   |            |                  |         |                          |         |               |                    |                     |                 |                              |
| Template          | TA Method                                                                                   |                |                                                                                                |            |                  |         | HFF Readout Mode         |         |               |                    |                     |                 |                              |
|                   | NONE                                                                                        |                |                                                                                                |            |                  |         | false                    |         |               |                    |                     |                 |                              |
| Dithers           | #                                                                                           | Dither Type    |                                                                                                |            | Size             |         | Starting Point           |         |               | Number of Points   |                     | Points          |                              |
|                   | 1                                                                                           | 4-POINT-DITHER |                                                                                                |            |                  |         |                          |         |               |                    |                     |                 |                              |
| Spectral Elements | #                                                                                           | Grating/Filter | Readout Pattern                                                                                | Groups/Int | Integrations/Exp | Leakcal | Dither                   | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |                              |
|                   | 1                                                                                           | G140H/F100LP   | NRSIRS2RAPID                                                                                   | 25         | 4                | false   | true                     | NONE    | 4             | 16                 | 6068.978            |                 |                              |
| Spectral Elements | 2                                                                                           | G235H/F170LP   | NRSIRS2RAPID                                                                                   | 12         | 3                | false   | true                     | NONE    | 4             | 12                 | 2275.867            |                 |                              |
|                   | 3                                                                                           | G395H/F290LP   | NRSIRS2RAPID                                                                                   | 7          | 3                | false   | true                     | NONE    | 4             | 12                 | 1400.533            |                 |                              |

# Proposal 12316 - Observation 2 - Probing the UV to MIR spectra of local Little Red Dots

|                   |                                                                                             |                |                                                                                                                                                                                                                                                  |            |                  |         |                          |         |                  |                    |                     |                              |
|-------------------|---------------------------------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|---------|--------------------------|---------|------------------|--------------------|---------------------|------------------------------|
| Observation       | Proposal 12316, Observation 2: J1025 NIRSPEC IFU                                            |                |                                                                                                                                                                                                                                                  |            |                  |         |                          |         |                  |                    |                     | Tue May 26 16:00:29 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                  |                |                                                                                                                                                                                                                                                  |            |                  |         |                          |         |                  |                    |                     |                              |
|                   | Observing Template: NIRSpec IFU Spectroscopy                                                |                |                                                                                                                                                                                                                                                  |            |                  |         |                          |         |                  |                    |                     |                              |
| Diagnostics       | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                |                                                                                                                                                                                                                                                  |            |                  |         |                          |         |                  |                    |                     |                              |
| Fixed Targets     | #                                                                                           | Name           | Target Coordinates                                                                                                                                                                                                                               |            |                  |         | Targ. Coord. Corrections |         |                  | Miscellaneous      |                     |                              |
|                   | (2)                                                                                         | SDSSJ1025+1402 | RA: 10 25 30.2919 (156.3762162d)                                                                                                                                                                                                                 |            |                  |         | Parallax: 0"             |         |                  |                    |                     |                              |
|                   |                                                                                             |                | Dec: +14 02 7.40 (14.03539d)                                                                                                                                                                                                                     |            |                  |         | Epoch of Position: 2016  |         |                  |                    |                     |                              |
|                   |                                                                                             |                | Equinox: J2000                                                                                                                                                                                                                                   |            |                  |         |                          |         |                  |                    |                     |                              |
|                   |                                                                                             |                | Comments: This object was generated by the targetselector and retrieved from the NED database.<br>Extent is ~0.5"x1"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES |            |                  |         |                          |         |                  |                    |                     |                              |
| Template          | TA Method                                                                                   |                |                                                                                                                                                                                                                                                  |            |                  |         | HFF Readout Mode         |         |                  |                    |                     |                              |
|                   | NONE                                                                                        |                |                                                                                                                                                                                                                                                  |            |                  |         | false                    |         |                  |                    |                     |                              |
| Dithers           | #                                                                                           | Dither Type    |                                                                                                                                                                                                                                                  |            | Size             |         | Starting Point           |         | Number of Points |                    | Points              |                              |
|                   | 1                                                                                           | 4-POINT-DITHER |                                                                                                                                                                                                                                                  |            |                  |         |                          |         |                  |                    |                     |                              |
| Spectral Elements | #                                                                                           | Grating/Filter | Readout Pattern                                                                                                                                                                                                                                  | Groups/Int | Integrations/Exp | Leakcal | Dither                   | Autocal | Total Dithers    | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                           | G140H/F100LP   | NRSIRS2RAPID                                                                                                                                                                                                                                     | 8          | 4                | false   | true                     | NONE    | 4                | 16                 | 2100.8              |                              |
|                   | 2                                                                                           | G235H/F170LP   | NRSIRS2RAPID                                                                                                                                                                                                                                     | 5          | 4                | false   | true                     | NONE    | 4                | 16                 | 1400.533            |                              |
|                   | 3                                                                                           | G395H/F290LP   | NRSIRS2RAPID                                                                                                                                                                                                                                     | 4          | 4                | false   | true                     | NONE    | 4                | 16                 | 1167.111            |                              |

# Proposal 12316 - Observation 3 - Probing the UV to MIR spectra of local Little Red Dots

|                                              |                                                                                                                                                                                                                                                                                                 |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|------------|------------------|----------------|--------------------------|------------------|---------------|--------------------|---------------------|------------------------------|
| Observation                                  | Proposal 12316, Observation 3: J1047 NIRSPEC IFU                                                                                                                                                                                                                                                |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     | Tue May 26 16:00:29 GMT 2026 |
|                                              | Diagnostic Status: Warning                                                                                                                                                                                                                                                                      |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Observing Template: NIRSpec IFU Spectroscopy |                                                                                                                                                                                                                                                                                                 |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Diagnostics                                  | (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                                     |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Fixed Targets                                | #                                                                                                                                                                                                                                                                                               | Name           | Target Coordinates                                                                 |            |                  |                | Targ. Coord. Corrections |                  |               | Miscellaneous      |                     |                              |
|                                              | (3)                                                                                                                                                                                                                                                                                             | SDSSJ1047+0739 | RA: 10 47 55.9266 (161.9830275d)<br>Dec: +07 39 51.23 (7.66423d)<br>Equinox: J2000 |            |                  |                | Epoch of Position: 2016  |                  |               |                    |                     |                              |
|                                              | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.                                                                                                                                                                                               |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
|                                              | SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
|                                              | Extragalactic objec at z=0.168. PM not relevant. Extent is ~0.5"x1.0"                                                                                                                                                                                                                           |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
|                                              | Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES                                                                                                                                                                        |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Template                                     | TA Method                                                                                                                                                                                                                                                                                       |                |                                                                                    |            |                  |                | HFF Readout Mode         |                  |               |                    |                     |                              |
|                                              | NONE                                                                                                                                                                                                                                                                                            |                |                                                                                    |            |                  |                | false                    |                  |               |                    |                     |                              |
| Dithers                                      | #                                                                                                                                                                                                                                                                                               | Dither Type    |                                                                                    | Size       |                  | Starting Point |                          | Number of Points |               | Points             |                     |                              |
|                                              | 1                                                                                                                                                                                                                                                                                               | 4-POINT-DITHER |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Spectral Elements                            | #                                                                                                                                                                                                                                                                                               | Grating/Filter | Readout Pattern                                                                    | Groups/Int | Integrations/Exp | Leakcal        | Dither                   | Autocal          | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                                              | 1                                                                                                                                                                                                                                                                                               | G140H/F100LP   | NRSIRS2RAPID                                                                       | 10         | 4                | false          | true                     | NONE             | 4             | 16                 | 2567.645            | 277379                       |
|                                              | 2                                                                                                                                                                                                                                                                                               | G235H/F170LP   | NRSIRS2RAPID                                                                       | 7          | 3                | false          | true                     | NONE             | 4             | 12                 | 1400.533            | 277379                       |
|                                              | 3                                                                                                                                                                                                                                                                                               | G395H/F290LP   | NRSIRS2RAPID                                                                       | 4          | 3                | false          | true                     | NONE             | 4             | 12                 | 875.333             | 277379                       |

# Proposal 12316 - Observation 4 - Probing the UV to MIR spectra of local Little Red Dots

|                                              |                                                                                                                          |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|------------|------------------|----------------|--------------------------|------------------|---------------|--------------------|---------------------|------------------------------|
| Observation                                  | Proposal 12316, Observation 4: J1340 NIRSPEC IFU                                                                         |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     | Tue May 26 16:00:29 GMT 2026 |
|                                              | Diagnostic Status: Warning                                                                                               |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Observing Template: NIRSpec IFU Spectroscopy |                                                                                                                          |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Diagnostics                                  | (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                              |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Fixed Targets                                | #                                                                                                                        | Name           | Target Coordinates                                                                 |            |                  |                | Targ. Coord. Corrections |                  |               | Miscellaneous      |                     |                              |
|                                              | (4)                                                                                                                      | SDSSJ1340+0211 | RA: 13 40 52.0334 (205.2168058d)<br>Dec: +02 11 57.27 (2.19924d)<br>Equinox: J2000 |            |                  |                | Epoch of Position: 2016  |                  |               |                    |                     |                              |
|                                              | Comments: This object was generated by the targetselector and retrieved from the NED database.<br>Extent is ~1-2"        |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
|                                              | Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Template                                     | TA Method                                                                                                                |                |                                                                                    |            |                  |                | HFF Readout Mode         |                  |               |                    |                     |                              |
|                                              | NONE                                                                                                                     |                |                                                                                    |            |                  |                | false                    |                  |               |                    |                     |                              |
| Dithers                                      | #                                                                                                                        | Dither Type    |                                                                                    | Size       |                  | Starting Point |                          | Number of Points |               | Points             |                     |                              |
|                                              | 1                                                                                                                        | 4-POINT-DITHER |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |
| Spectral Elements                            | #                                                                                                                        | Grating/Filter | Readout Pattern                                                                    | Groups/Int | Integrations/Exp | Leakcal        | Dither                   | Autocal          | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                                              | 1                                                                                                                        | G140H/F100LP   | NRSIRS2RAPID                                                                       | 5          | 3                | false          | true                     | NONE             | 4             | 12                 | 1050.4              | 277379                       |
|                                              | 2                                                                                                                        | G235H/F170LP   | NRSIRS2RAPID                                                                       | 4          | 3                | false          | true                     | NONE             | 4             | 12                 | 875.333             | 277379                       |
|                                              | 3                                                                                                                        | G395H/F290LP   | NRSIRS2RAPID                                                                       | 5          | 3                | false          | true                     | NONE             | 4             | 12                 | 1050.4              | 277379                       |
| Special Requirements                         | Aperture PA Range 184.97164917 to 72.97164917 Degrees (V3 46.0 to 294.0)                                                 |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                              |

# Proposal 12316 - Observation 5 - Probing the UV to MIR spectra of local Little Red Dots

|                                              |                                                                                                                                                                                                                                      |                |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                              |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|------------|------------------|----------------|--------------------------|------------------|---------------|--------------------|---------------------|------------------------------|
| Observation                                  | Proposal 12316, Observation 5: J1717 NIRSPEC IFU                                                                                                                                                                                     |                |                                                                                     |            |                  |                |                          |                  |               |                    |                     | Tue May 26 16:00:29 GMT 2026 |
|                                              | Diagnostic Status: Warning                                                                                                                                                                                                           |                |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                              |
| Observing Template: NIRSpec IFU Spectroscopy |                                                                                                                                                                                                                                      |                |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                              |
| Diagnostics                                  | (Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                          |                |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                              |
| Fixed Targets                                | #                                                                                                                                                                                                                                    | Name           | Target Coordinates                                                                  |            |                  |                | Targ. Coord. Corrections |                  |               | Miscellaneous      |                     |                              |
|                                              | (5)                                                                                                                                                                                                                                  | SDSSJ1717+3807 | RA: 17 17 41.7363 (259.4239013d)<br>Dec: +38 07 52.47 (38.13124d)<br>Equinox: J2000 |            |                  |                | Epoch of Position: 2016  |                  |               |                    |                     |                              |
|                                              | Comments: This object was generated by the targetselector and retrieved from the NED database.                                                                                                                                       |                |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                              |
|                                              | Extent is <1"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES                                                                                            |                |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                              |
| Template                                     | TA Method                                                                                                                                                                                                                            |                |                                                                                     |            |                  |                | HFF Readout Mode         |                  |               |                    |                     |                              |
|                                              | NONE                                                                                                                                                                                                                                 |                |                                                                                     |            |                  |                | false                    |                  |               |                    |                     |                              |
| Dithers                                      | #                                                                                                                                                                                                                                    | Dither Type    |                                                                                     | Size       |                  | Starting Point |                          | Number of Points |               | Points             |                     |                              |
|                                              | 1                                                                                                                                                                                                                                    | 4-POINT-DITHER |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                              |
| Spectral Elements                            | #                                                                                                                                                                                                                                    | Grating/Filter | Readout Pattern                                                                     | Groups/Int | Integrations/Exp | Leakcal        | Dither                   | Autocal          | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                                              | 1                                                                                                                                                                                                                                    | G140H/F100LP   | NRSIRS2RAPID                                                                        | 5          | 3                | false          | true                     | NONE             | 4             | 12                 | 1050.4              | 227213                       |
|                                              | 2                                                                                                                                                                                                                                    | G235H/F170LP   | NRSIRS2RAPID                                                                        | 3          | 3                | false          | true                     | NONE             | 4             | 12                 | 700.267             | 227213                       |
|                                              | 3                                                                                                                                                                                                                                    | G395H/F290LP   | NRSIRS2RAPID                                                                        | 4          | 2                | false          | true                     | NONE             | 4             | 8                  | 583.556             | 227213                       |
| Special Requirements                         | Aperture PA Range 138.97164917 to 227.97164917 Degrees (V3 0.0 to 89.0)<br>Aperture PA Range 240.97164917 to 263.97164917 Degrees (V3 102.0 to 125.0)<br>Aperture PA Range 276.97164917 to 138.96164917 Degrees (V3 138.0 to 359.99) |                |                                                                                     |            |                  |                |                          |                  |               |                    |                     |                              |



# Proposal 12316 - Observation 6 - Probing the UV to MIR spectra of local Little Red Dots

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |                                                                                                                                                                                                                                                                                           |            |                  |                |                          |                  |               |                    |                     |                              |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|----------------|--------------------------|------------------|---------------|--------------------|---------------------|------------------------------|
| Observation          | Proposal 12316, Observation 6: TOL2240 NIRSPEC IFU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |                                                                                                                                                                                                                                                                                           |            |                  |                |                          |                  |               |                    |                     | Tue May 26 16:00:29 GMT 2026 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                |                                                                                                                                                                                                                                                                                           |            |                  |                |                          |                  |               |                    |                     |                              |
|                      | Observing Template: NIRSpec IFU Spectroscopy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |                                                                                                                                                                                                                                                                                           |            |                  |                |                          |                  |               |                    |                     |                              |
| Diagnostics          | (Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                |                                                                                                                                                                                                                                                                                           |            |                  |                |                          |                  |               |                    |                     |                              |
| Fixed Targets        | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Name           | Target Coordinates                                                                                                                                                                                                                                                                        |            |                  |                | Targ. Coord. Corrections |                  |               | Miscellaneous      |                     |                              |
|                      | (6)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | TOL2240-384    | RA: 22 43 32.9629 (340.8873454d)                                                                                                                                                                                                                                                          |            |                  |                | Epoch of Position: 2016  |                  |               |                    |                     |                              |
|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                | Dec: -38 11 20.74 (-38.18909d)                                                                                                                                                                                                                                                            |            |                  |                |                          |                  |               |                    |                     |                              |
|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                | Equinox: J2000                                                                                                                                                                                                                                                                            |            |                  |                |                          |                  |               |                    |                     |                              |
|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                | Comments: Extent is 1"x2.5". Pointing center shifted NE by ~0.3" relative to peak, so IFU will cover point source and NE extension (based on LS DR10 imaging)<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES |            |                  |                |                          |                  |               |                    |                     |                              |
| Template             | TA Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                |                                                                                                                                                                                                                                                                                           |            |                  |                | HFF Readout Mode         |                  |               |                    |                     |                              |
|                      | NONE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |                                                                                                                                                                                                                                                                                           |            |                  |                | false                    |                  |               |                    |                     |                              |
| Dithers              | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Dither Type    |                                                                                                                                                                                                                                                                                           | Size       |                  | Starting Point |                          | Number of Points |               | Points             |                     |                              |
|                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4-POINT-DITHER |                                                                                                                                                                                                                                                                                           |            |                  |                |                          |                  |               |                    |                     |                              |
| Spectral Elements    | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Grating/Filter | Readout Pattern                                                                                                                                                                                                                                                                           | Groups/Int | Integrations/Exp | Leakcal        | Dither                   | Autocal          | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | G140H/F100LP   | NRSIRS2RAPID                                                                                                                                                                                                                                                                              | 9          | 3                | false          | true                     | NONE             | 4             | 12                 | 1750.667            | 227213                       |
|                      | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | G235H/F170LP   | NRSIRS2RAPID                                                                                                                                                                                                                                                                              | 4          | 3                | false          | true                     | NONE             | 4             | 12                 | 875.333             | 227213                       |
|                      | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | G395H/F290LP   | NRSIRS2RAPID                                                                                                                                                                                                                                                                              | 4          | 2                | false          | true                     | NONE             | 4             | 8                  | 583.556             | 227213                       |
| Special Requirements | Aperture PA Range 82.97164917 to 97.97164917 Degrees (V3 304.0 to 319.0)<br>Aperture PA Range 111.97164917 to 138.97064917 Degrees (V3 333.0 to 359.999)<br>Aperture PA Range 138.97164917 to 166.97164917 Degrees (V3 0.0 to 28.0)<br>Aperture PA Range 177.97164917 to 208.97164917 Degrees (V3 39.0 to 70.0)<br>Aperture PA Range 233.97164917 to 268.97164917 Degrees (V3 95.0 to 130.0)<br>Aperture PA Range 318.97164917 to 341.97164917 Degrees (V3 180.0 to 203.0)<br>Aperture PA Range 356.97164917 to 31.97164917 Degrees (V3 218.0 to 253.0) |                |                                                                                                                                                                                                                                                                                           |            |                  |                |                          |                  |               |                    |                     |                              |

# Proposal 12316 - Observation 7 - Probing the UV to MIR spectra of local Little Red Dots

|                   |                                                                                             |                                                                                                                                                                                                                                             |          |        |                                 |                      |                  |                |                          |                 |                    |                     |                 |                              |  |  |  |
|-------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|---------------------------------|----------------------|------------------|----------------|--------------------------|-----------------|--------------------|---------------------|-----------------|------------------------------|--|--|--|
| Observation       | Proposal 12316, Observation 7: J1022 MIRI MRS                                               |                                                                                                                                                                                                                                             |          |        |                                 |                      |                  |                |                          |                 |                    |                     |                 | Tue May 26 16:00:29 GMT 2026 |  |  |  |
|                   | Diagnostic Status: Warning                                                                  |                                                                                                                                                                                                                                             |          |        |                                 |                      |                  |                |                          |                 |                    |                     |                 |                              |  |  |  |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                     |                                                                                                                                                                                                                                             |          |        |                                 |                      |                  |                |                          |                 |                    |                     |                 |                              |  |  |  |
| Diagnostics       | (Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                                                                                                                                                                             |          |        |                                 |                      |                  |                |                          |                 |                    |                     |                 |                              |  |  |  |
| Fixed Targets     | #                                                                                           | Name                                                                                                                                                                                                                                        |          |        | Target Coordinates              |                      |                  |                | Targ. Coord. Corrections |                 |                    |                     | Miscellaneous   |                              |  |  |  |
|                   | (1)                                                                                         | SDSSJ1022+0841                                                                                                                                                                                                                              |          |        | RA: 10 22 8.5286 (155.5355358d) |                      |                  |                | Parallax: 0"             |                 |                    |                     |                 |                              |  |  |  |
|                   |                                                                                             |                                                                                                                                                                                                                                             |          |        | Dec: +08 41 56.19 (8.69894d)    |                      |                  |                | Epoch of Position: 2016  |                 |                    |                     |                 |                              |  |  |  |
|                   |                                                                                             |                                                                                                                                                                                                                                             |          |        | Equinox: J2000                  |                      |                  |                |                          |                 |                    |                     |                 |                              |  |  |  |
|                   |                                                                                             | Comments: This object was generated by the targetselector and retrieved from the NED database.<br>Extent is <1"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES |          |        |                                 |                      |                  |                |                          |                 |                    |                     |                 |                              |  |  |  |
| Acquisition       | #                                                                                           | Target                                                                                                                                                                                                                                      |          |        |                                 |                      |                  |                |                          |                 |                    |                     |                 |                              |  |  |  |
|                   | 1                                                                                           | NONE                                                                                                                                                                                                                                        |          |        |                                 |                      |                  |                |                          |                 |                    |                     |                 |                              |  |  |  |
| Template          | AcqFilter                                                                                   | Primary Channel                                                                                                                                                                                                                             |          |        |                                 | Simultaneous Imaging |                  |                |                          | Imager Subarray |                    |                     |                 | Grating Wheel Direction      |  |  |  |
|                   |                                                                                             | All MRS                                                                                                                                                                                                                                     |          |        |                                 | NO                   |                  |                |                          | FULL            |                    |                     |                 | Allow Auto Reorder           |  |  |  |
| Dithers           | #                                                                                           | Dither Type                                                                                                                                                                                                                                 |          |        |                                 | Optimized For        |                  |                |                          | Direction       |                    |                     |                 |                              |  |  |  |
|                   | 1                                                                                           | 4-Point                                                                                                                                                                                                                                     |          |        |                                 | EXTENDED SOURCE      |                  |                |                          | NEGATIVE        |                    |                     |                 |                              |  |  |  |
| Spectral Elements | #                                                                                           | Wavelength Range                                                                                                                                                                                                                            | Detector | Filter | Readout Pattern                 | Groups/Int           | Integrations/Exp | Exposures/Dith | Dither                   | Total Dithers   | Total Integrations | Total Exposure Time | Optional ETC ID |                              |  |  |  |
|                   | 1                                                                                           | SHORT(A)                                                                                                                                                                                                                                    | MRSLONG  |        | FASTR1                          | 100                  | 3                | 1              | Dither 1                 | 4               | 12                 | 3352.248            | 305707          |                              |  |  |  |
|                   | 1                                                                                           | SHORT(A)                                                                                                                                                                                                                                    | MRSSHORT |        | FASTR1                          | 100                  | 3                | 1              | Dither 1                 | 4               | 12                 | 3352.248            | 305707          |                              |  |  |  |
|                   | 2                                                                                           | MEDIUM(B)                                                                                                                                                                                                                                   | MRSLONG  |        | FASTR1                          | 60                   | 3                | 1              | Dither 1                 | 4               | 12                 | 2020.229            | 305707          |                              |  |  |  |
|                   | 2                                                                                           | MEDIUM(B)                                                                                                                                                                                                                                   | MRSSHORT |        | FASTR1                          | 60                   | 3                | 1              | Dither 1                 | 4               | 12                 | 2020.229            | 305707          |                              |  |  |  |
|                   | 3                                                                                           | LONG(C)                                                                                                                                                                                                                                     | MRSLONG  |        | FASTR1                          | 40                   | 3                | 1              | Dither 1                 | 4               | 12                 | 1354.22             | 305707          |                              |  |  |  |
|                   | 3                                                                                           | LONG(C)                                                                                                                                                                                                                                     | MRSSHORT |        | FASTR1                          | 40                   | 3                | 1              | Dither 1                 | 4               | 12                 | 1354.22             | 305707          |                              |  |  |  |
|                   |                                                                                             |                                                                                                                                                                                                                                             |          |        |                                 |                      |                  |                |                          |                 |                    |                     |                 |                              |  |  |  |

# Proposal 12316 - Observation 8 - Probing the UV to MIR spectra of local Little Red Dots

|                   |                                                                                                                                                                                                                                                  |                  |                                                                                    |        |                 |                      |                                         |                |          |                 |                    |                         |                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------|--------|-----------------|----------------------|-----------------------------------------|----------------|----------|-----------------|--------------------|-------------------------|------------------------------|
| Observation       | Proposal 12316, Observation 8: J1025 MIRI MRS                                                                                                                                                                                                    |                  |                                                                                    |        |                 |                      |                                         |                |          |                 |                    |                         | Tue May 26 16:00:29 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                                       |                  |                                                                                    |        |                 |                      |                                         |                |          |                 |                    |                         |                              |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                                                                                                                                                                          |                  |                                                                                    |        |                 |                      |                                         |                |          |                 |                    |                         |                              |
| Diagnostics       | (Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                      |                  |                                                                                    |        |                 |                      |                                         |                |          |                 |                    |                         |                              |
| Fixed Targets     | #                                                                                                                                                                                                                                                | Name             | Target Coordinates                                                                 |        |                 |                      | Targ. Coord. Corrections                |                |          |                 | Miscellaneous      |                         |                              |
|                   | (2)                                                                                                                                                                                                                                              | SDSSJ1025+1402   | RA: 10 25 30.2919 (156.3762162d)<br>Dec: +14 02 7.40 (14.03539d)<br>Equinox: J2000 |        |                 |                      | Parallax: 0"<br>Epoch of Position: 2016 |                |          |                 |                    |                         |                              |
|                   | Comments: This object was generated by the targetselector and retrieved from the NED database.<br>Extent is ~0.5"x1"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES |                  |                                                                                    |        |                 |                      |                                         |                |          |                 |                    |                         |                              |
| Acquisition       | #                                                                                                                                                                                                                                                | Target           |                                                                                    |        |                 |                      |                                         |                |          |                 |                    |                         |                              |
|                   | 1                                                                                                                                                                                                                                                | NONE             |                                                                                    |        |                 |                      |                                         |                |          |                 |                    |                         |                              |
| Template          | AcqFilter                                                                                                                                                                                                                                        | Primary Channel  |                                                                                    |        |                 | Simultaneous Imaging |                                         |                |          | Imager Subarray |                    | Grating Wheel Direction |                              |
|                   |                                                                                                                                                                                                                                                  | All MRS          |                                                                                    |        |                 | NO                   |                                         |                |          | FULL            |                    | Allow Auto Reorder      |                              |
| Dithers           | #                                                                                                                                                                                                                                                | Dither Type      |                                                                                    |        |                 | Optimized For        |                                         |                |          | Direction       |                    |                         |                              |
|                   | 1                                                                                                                                                                                                                                                | 4-Point          |                                                                                    |        |                 | EXTENDED SOURCE      |                                         |                |          | NEGATIVE        |                    |                         |                              |
| Spectral Elements | #                                                                                                                                                                                                                                                | Wavelength Range | Detector                                                                           | Filter | Readout Pattern | Groups/Int           | Integrations/Exp                        | Exposures/Dith | Dither   | Total Dithers   | Total Integrations | Total Exposure Time     | Optional ETC ID              |
|                   | 1                                                                                                                                                                                                                                                | SHORT(A)         | MRSLONG                                                                            |        | FASTR1          | 85                   | 3                                       | 1              | Dither 1 | 4               | 12                 | 2852.741                | 305707                       |
|                   | 1                                                                                                                                                                                                                                                | SHORT(A)         | MRSSHORT                                                                           |        | FASTR1          | 85                   | 3                                       | 1              | Dither 1 | 4               | 12                 | 2852.741                | 305707                       |
|                   | 2                                                                                                                                                                                                                                                | MEDIUM(B)        | MRSLONG                                                                            |        | FASTR1          | 60                   | 3                                       | 1              | Dither 1 | 4               | 12                 | 2020.229                | 305707                       |
|                   | 2                                                                                                                                                                                                                                                | MEDIUM(B)        | MRSSHORT                                                                           |        | FASTR1          | 60                   | 3                                       | 1              | Dither 1 | 4               | 12                 | 2020.229                | 305707                       |
|                   | 3                                                                                                                                                                                                                                                | LONG(C)          | MRSLONG                                                                            |        | FASTR1          | 45                   | 3                                       | 1              | Dither 1 | 4               | 12                 | 1520.722                | 305707                       |
|                   | 3                                                                                                                                                                                                                                                | LONG(C)          | MRSSHORT                                                                           |        | FASTR1          | 45                   | 3                                       | 1              | Dither 1 | 4               | 12                 | 1520.722                | 305707                       |

# Proposal 12316 - Observation 9 - Probing the UV to MIR spectra of local Little Red Dots

|                   |                                                                                                                                                                                                                                                                                                 |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------|--------|-----------------|----------------------|--------------------------|----------------|----------|-----------------|--------------------|---------------------|-------------------------|------------------------------|
| Observation       | Proposal 12316, Observation 9: J1047 MIRI MRS                                                                                                                                                                                                                                                   |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         | Tue May 26 16:00:29 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                                                                                      |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                                                                                                                                                                                                                         |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
| Diagnostics       | (Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                                     |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
| Fixed Targets     | #                                                                                                                                                                                                                                                                                               | Name             | Target Coordinates                                                                 |        |                 |                      | Targ. Coord. Corrections |                |          |                 | Miscellaneous      |                     |                         |                              |
|                   | (3)                                                                                                                                                                                                                                                                                             | SDSSJ1047+0739   | RA: 10 47 55.9266 (161.9830275d)<br>Dec: +07 39 51.23 (7.66423d)<br>Equinox: J2000 |        |                 |                      | Epoch of Position: 2016  |                |          |                 |                    |                     |                         |                              |
|                   | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.                                                                                                                                                                                               |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
|                   | SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
|                   | Extragalactic objec at z=0.168. PM not relevant. Extent is ~0.5"x1.0"                                                                                                                                                                                                                           |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
|                   | Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES                                                                                                                                                                        |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
| Acquisition       | #                                                                                                                                                                                                                                                                                               | Target           |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
|                   | 1                                                                                                                                                                                                                                                                                               | NONE             |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |
| Template          | AcqFilter                                                                                                                                                                                                                                                                                       | Primary Channel  |                                                                                    |        |                 | Simultaneous Imaging |                          |                |          | Imager Subarray |                    |                     | Grating Wheel Direction |                              |
|                   |                                                                                                                                                                                                                                                                                                 | All MRS          |                                                                                    |        |                 | NO                   |                          |                |          | FULL            |                    |                     | Allow Auto Reorder      |                              |
| Dithers           | #                                                                                                                                                                                                                                                                                               | Dither Type      |                                                                                    |        |                 | Optimized For        |                          |                |          | Direction       |                    |                     |                         |                              |
|                   | 1                                                                                                                                                                                                                                                                                               | 4-Point          |                                                                                    |        |                 | EXTENDED SOURCE      |                          |                |          | NEGATIVE        |                    |                     |                         |                              |
| Spectral Elements | #                                                                                                                                                                                                                                                                                               | Wavelength Range | Detector                                                                           | Filter | Readout Pattern | Groups/Int           | Integrations/Exp         | Exposures/Dith | Dither   | Total Dithers   | Total Integrations | Total Exposure Time | Optional ETC ID         |                              |
|                   | 1                                                                                                                                                                                                                                                                                               | SHORT(A)         | MRSLONG                                                                            |        | FASTR1          | 100                  | 3                        | 1              | Dither 1 | 4               | 12                 | 3352.248            | 305707                  |                              |
|                   | 1                                                                                                                                                                                                                                                                                               | SHORT(A)         | MRSSHORT                                                                           |        | FASTR1          | 100                  | 3                        | 1              | Dither 1 | 4               | 12                 | 3352.248            | 305707                  |                              |
|                   | 2                                                                                                                                                                                                                                                                                               | MEDIUM(B)        | MRSLONG                                                                            |        | FASTR1          | 76                   | 3                        | 1              | Dither 1 | 4               | 12                 | 2553.037            | 305707                  |                              |
|                   | 2                                                                                                                                                                                                                                                                                               | MEDIUM(B)        | MRSSHORT                                                                           |        | FASTR1          | 76                   | 3                        | 1              | Dither 1 | 4               | 12                 | 2553.037            | 305707                  |                              |
|                   | 3                                                                                                                                                                                                                                                                                               | LONG(C)          | MRSLONG                                                                            |        | FASTR1          | 42                   | 3                        | 1              | Dither 1 | 4               | 12                 | 1420.82             | 305707                  |                              |
|                   | 3                                                                                                                                                                                                                                                                                               | LONG(C)          | MRSSHORT                                                                           |        | FASTR1          | 42                   | 3                        | 1              | Dither 1 | 4               | 12                 | 1420.82             | 305707                  |                              |
|                   |                                                                                                                                                                                                                                                                                                 |                  |                                                                                    |        |                 |                      |                          |                |          |                 |                    |                     |                         |                              |

# Proposal 12316 - Observation 10 - Probing the UV to MIR spectra of local Little Red Dots

|                   |                                                                                              |                  |                                                                                    |        |                 |                      |                  |                          |                 |               |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |
|-------------------|----------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------|--------|-----------------|----------------------|------------------|--------------------------|-----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|------------------------------|--|--|
| Observation       | Proposal 12316, Observation 10: J1340 MIRI MRS                                               |                  |                                                                                    |        |                 |                      |                  |                          |                 |               |                                                                                                                                                                                                                                                                                  |                     |                 | Tue May 26 16:00:29 GMT 2026 |  |  |
|                   | Diagnostic Status: Warning                                                                   |                  |                                                                                    |        |                 |                      |                  |                          |                 |               |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                      |                  |                                                                                    |        |                 |                      |                  |                          |                 |               |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |
| Diagnostics       | (Visit 10:1) Warning (Form): Data Excess over lower threshold                                |                  |                                                                                    |        |                 |                      |                  |                          |                 |               |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |
|                   | (Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                  |                                                                                    |        |                 |                      |                  |                          |                 |               |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |
| Fixed Targets     | #                                                                                            | Name             | Target Coordinates                                                                 |        |                 |                      |                  | Targ. Coord. Corrections |                 |               | Miscellaneous                                                                                                                                                                                                                                                                    |                     |                 |                              |  |  |
|                   | (4)                                                                                          | SDSSJ1340+0211   | RA: 13 40 52.0334 (205.2168058d)<br>Dec: +02 11 57.27 (2.19924d)<br>Equinox: J2000 |        |                 |                      |                  | Epoch of Position: 2016  |                 |               | <i>Comments: This object was generated by the targetselector and retrieved from the NED database.</i><br><i>Extent is ~1-2"</i><br><i>Category=Galaxy</i><br><i>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]</i><br><i>Extended=YES</i> |                     |                 |                              |  |  |
| Acquisition       | #                                                                                            | Target           |                                                                                    |        |                 |                      |                  |                          |                 |               |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |
|                   | 1                                                                                            | NONE             |                                                                                    |        |                 |                      |                  |                          |                 |               |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |
| Template          | AcqFilter                                                                                    | Primary Channel  |                                                                                    |        |                 | Simultaneous Imaging |                  |                          | Imager Subarray |               |                                                                                                                                                                                                                                                                                  |                     |                 | Grating Wheel Direction      |  |  |
|                   |                                                                                              | All MRS          |                                                                                    |        |                 | NO                   |                  |                          | FULL            |               | Allow Auto Reorder                                                                                                                                                                                                                                                               |                     |                 |                              |  |  |
| Dithers           | #                                                                                            | Dither Type      |                                                                                    |        |                 |                      | Optimized For    |                          |                 | Direction     |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |
|                   | 1                                                                                            | 4-Point          |                                                                                    |        |                 |                      | EXTENDED SOURCE  |                          |                 | NEGATIVE      |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |
| Spectral Elements | #                                                                                            | Wavelength Range | Detector                                                                           | Filter | Readout Pattern | Groups/Int           | Integrations/Exp | Exposures/Dith           | Dither          | Total Dithers | Total Integrations                                                                                                                                                                                                                                                               | Total Exposure Time | Optional ETC ID |                              |  |  |
|                   | 1                                                                                            | SHORT(A)         | MRSLONG                                                                            |        | FASTR1          | 112                  | 3                | 1                        | Dither 1        | 4             | 12                                                                                                                                                                                                                                                                               | 3751.854            | 305706          |                              |  |  |
|                   | 1                                                                                            | SHORT(A)         | MRSSHORT                                                                           |        | FASTR1          | 112                  | 3                | 1                        | Dither 1        | 4             | 12                                                                                                                                                                                                                                                                               | 3751.854            | 305706          |                              |  |  |
|                   | 2                                                                                            | MEDIUM(B)        | MRSLONG                                                                            |        | FASTR1          | 90                   | 3                | 1                        | Dither 1        | 4             | 12                                                                                                                                                                                                                                                                               | 3019.244            | 305706          |                              |  |  |
|                   | 2                                                                                            | MEDIUM(B)        | MRSSHORT                                                                           |        | FASTR1          | 90                   | 3                | 1                        | Dither 1        | 4             | 12                                                                                                                                                                                                                                                                               | 3019.244            | 305706          |                              |  |  |
|                   | 3                                                                                            | LONG(C)          | MRSLONG                                                                            |        | FASTR1          | 60                   | 3                | 1                        | Dither 1        | 4             | 12                                                                                                                                                                                                                                                                               | 2020.229            | 305706          |                              |  |  |
|                   | 3                                                                                            | LONG(C)          | MRSSHORT                                                                           |        | FASTR1          | 60                   | 3                | 1                        | Dither 1        | 4             | 12                                                                                                                                                                                                                                                                               | 2020.229            | 305706          |                              |  |  |
|                   |                                                                                              |                  |                                                                                    |        |                 |                      |                  |                          |                 |               |                                                                                                                                                                                                                                                                                  |                     |                 |                              |  |  |

# Proposal 12316 - Observation 11 - Probing the UV to MIR spectra of local Little Red Dots

|                   |                                                                                              |                  |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      |                  |                          |                 |               |                    |                         |                 |                              |
|-------------------|----------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|----------------------|------------------|--------------------------|-----------------|---------------|--------------------|-------------------------|-----------------|------------------------------|
| Observation       | Proposal 12316, Observation 11: J1717 MIRI MRS                                               |                  |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      |                  |                          |                 |               |                    |                         |                 | Tue May 26 16:00:29 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                   |                  |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      |                  |                          |                 |               |                    |                         |                 |                              |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                      |                  |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      |                  |                          |                 |               |                    |                         |                 |                              |
| Diagnostics       | (Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                  |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      |                  |                          |                 |               |                    |                         |                 |                              |
| Fixed Targets     | #                                                                                            | Name             | Target Coordinates                                                                                                                                                                                                                                                                                                                     |        |                 |                      |                  | Targ. Coord. Corrections |                 |               | Miscellaneous      |                         |                 |                              |
|                   | (5)                                                                                          | SDSSJ1717+3807   | RA: 17 17 41.7363 (259.4239013d)<br>Dec: +38 07 52.47 (38.13124d)<br>Equinox: J2000<br><br>Comments: This object was generated by the targetselector and retrieved from the NED database.<br>Extent is <1"<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES |        |                 |                      |                  | Epoch of Position: 2016  |                 |               |                    |                         |                 |                              |
| Acquisition       | #                                                                                            | Target           |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      |                  |                          |                 |               |                    |                         |                 |                              |
|                   | 1                                                                                            | NONE             |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      |                  |                          |                 |               |                    |                         |                 |                              |
| Template          | AcqFilter                                                                                    | Primary Channel  |                                                                                                                                                                                                                                                                                                                                        |        |                 | Simultaneous Imaging |                  |                          | Imager Subarray |               |                    | Grating Wheel Direction |                 |                              |
|                   |                                                                                              | All MRS          |                                                                                                                                                                                                                                                                                                                                        |        |                 | NO                   |                  |                          | FULL            |               |                    | Allow Auto Reorder      |                 |                              |
| Dithers           | #                                                                                            | Dither Type      |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      | Optimized For    |                          |                 | Direction     |                    |                         |                 |                              |
|                   | 1                                                                                            | 4-Point          |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      | EXTENDED SOURCE  |                          |                 | NEGATIVE      |                    |                         |                 |                              |
| Spectral Elements | #                                                                                            | Wavelength Range | Detector                                                                                                                                                                                                                                                                                                                               | Filter | Readout Pattern | Groups/Int           | Integrations/Exp | Exposures/Dith           | Dither          | Total Dithers | Total Integrations | Total Exposure Time     | Optional ETC ID |                              |
|                   | 1                                                                                            | SHORT(A)         | MRSLONG                                                                                                                                                                                                                                                                                                                                |        | FASTR1          | 24                   | 3                | 1                        | Dither 1        | 4             | 12                 | 821.412                 | 305706          |                              |
|                   | 1                                                                                            | SHORT(A)         | MRSSHORT                                                                                                                                                                                                                                                                                                                               |        | FASTR1          | 24                   | 3                | 1                        | Dither 1        | 4             | 12                 | 821.412                 | 305706          |                              |
|                   | 2                                                                                            | MEDIUM(B)        | MRSLONG                                                                                                                                                                                                                                                                                                                                |        | FASTR1          | 18                   | 3                | 1                        | Dither 1        | 4             | 12                 | 621.609                 | 305706          |                              |
|                   | 2                                                                                            | MEDIUM(B)        | MRSSHORT                                                                                                                                                                                                                                                                                                                               |        | FASTR1          | 18                   | 3                | 1                        | Dither 1        | 4             | 12                 | 621.609                 | 305706          |                              |
|                   | 3                                                                                            | LONG(C)          | MRSLONG                                                                                                                                                                                                                                                                                                                                |        | FASTR1          | 13                   | 3                | 1                        | Dither 1        | 4             | 12                 | 455.107                 | 305706          |                              |
|                   | 3                                                                                            | LONG(C)          | MRSSHORT                                                                                                                                                                                                                                                                                                                               |        | FASTR1          | 13                   | 3                | 1                        | Dither 1        | 4             | 12                 | 455.107                 | 305706          |                              |
|                   |                                                                                              |                  |                                                                                                                                                                                                                                                                                                                                        |        |                 |                      |                  |                          |                 |               |                    |                         |                 |                              |

# Proposal 12316 - Observation 12 - Probing the UV to MIR spectra of local Little Red Dots

|                   |                                                                                              |                                                                                                                                                                                                                                                                                           |                                  |        |                 |                      |                          |                 |          |                         |                    |                     |                              |
|-------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------|-----------------|----------------------|--------------------------|-----------------|----------|-------------------------|--------------------|---------------------|------------------------------|
| Observation       | Proposal 12316, Observation 12: Tol2240 MIRI MRS                                             |                                                                                                                                                                                                                                                                                           |                                  |        |                 |                      |                          |                 |          |                         |                    |                     | Tue May 26 16:00:29 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                   |                                                                                                                                                                                                                                                                                           |                                  |        |                 |                      |                          |                 |          |                         |                    |                     |                              |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                      |                                                                                                                                                                                                                                                                                           |                                  |        |                 |                      |                          |                 |          |                         |                    |                     |                              |
| Diagnostics       | (Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                                                                                                                                                                                                                           |                                  |        |                 |                      |                          |                 |          |                         |                    |                     |                              |
| Fixed Targets     | #                                                                                            | Name                                                                                                                                                                                                                                                                                      | Target Coordinates               |        |                 |                      | Targ. Coord. Corrections |                 |          | Miscellaneous           |                    |                     |                              |
|                   | (6)                                                                                          | TOL2240-384                                                                                                                                                                                                                                                                               | RA: 22 43 32.9629 (340.8873454d) |        |                 |                      | Epoch of Position: 2016  |                 |          |                         |                    |                     |                              |
|                   |                                                                                              |                                                                                                                                                                                                                                                                                           | Dec: -38 11 20.74 (-38.18909d)   |        |                 |                      |                          |                 |          |                         |                    |                     |                              |
|                   |                                                                                              |                                                                                                                                                                                                                                                                                           | Equinox: J2000                   |        |                 |                      |                          |                 |          |                         |                    |                     |                              |
|                   |                                                                                              | Comments: Extent is 1"x2.5". Pointing center shifted NE by ~0.3" relative to peak, so IFU will cover point source and NE extension (based on LS DR10 imaging)<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES |                                  |        |                 |                      |                          |                 |          |                         |                    |                     |                              |
| Acquisition       | #                                                                                            | Target                                                                                                                                                                                                                                                                                    |                                  |        |                 |                      |                          |                 |          |                         |                    |                     |                              |
|                   | 1                                                                                            | NONE                                                                                                                                                                                                                                                                                      |                                  |        |                 |                      |                          |                 |          |                         |                    |                     |                              |
| Template          | AcqFilter                                                                                    | Primary Channel                                                                                                                                                                                                                                                                           |                                  |        |                 | Simultaneous Imaging |                          | Imager Subarray |          | Grating Wheel Direction |                    |                     |                              |
|                   |                                                                                              | All MRS                                                                                                                                                                                                                                                                                   |                                  |        |                 | YES                  |                          | FULL            |          | Allow Auto Reorder      |                    |                     |                              |
| Dithers           | #                                                                                            | Dither Type                                                                                                                                                                                                                                                                               |                                  |        |                 | Optimized For        |                          |                 |          | Direction               |                    |                     |                              |
|                   | 1                                                                                            | 4-Point                                                                                                                                                                                                                                                                                   |                                  |        |                 | EXTENDED SOURCE      |                          |                 |          | NEGATIVE                |                    |                     |                              |
| Spectral Elements | #                                                                                            | Wavelength Range                                                                                                                                                                                                                                                                          | Detector                         | Filter | Readout Pattern | Groups/Int           | Integrations/Exp         | Exposures/Dith  | Dither   | Total Dithers           | Total Integrations | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                            |                                                                                                                                                                                                                                                                                           | IMAGER                           | F560W  | FASTR1          | 20                   | 3                        | 1               | Dither 1 | 4                       | 12                 | 688.21              |                              |
|                   | 1                                                                                            | SHORT(A)                                                                                                                                                                                                                                                                                  | MRSLONG                          |        | FASTR1          | 20                   | 3                        | 1               | Dither 1 | 4                       | 12                 | 688.21              | 305707                       |
|                   | 1                                                                                            | SHORT(A)                                                                                                                                                                                                                                                                                  | MRSSHORT                         |        | FASTR1          | 20                   | 3                        | 1               | Dither 1 | 4                       | 12                 | 688.21              | 305707                       |
|                   | 2                                                                                            |                                                                                                                                                                                                                                                                                           | IMAGER                           | F770W  | FASTR1          | 13                   | 3                        | 1               | Dither 1 | 4                       | 12                 | 455.107             |                              |
|                   | 2                                                                                            | MEDIUM(B)                                                                                                                                                                                                                                                                                 | MRSLONG                          |        | FASTR1          | 13                   | 3                        | 1               | Dither 1 | 4                       | 12                 | 455.107             | 305707                       |
|                   | 2                                                                                            | MEDIUM(B)                                                                                                                                                                                                                                                                                 | MRSSHORT                         |        | FASTR1          | 13                   | 3                        | 1               | Dither 1 | 4                       | 12                 | 455.107             | 305707                       |
|                   | 3                                                                                            |                                                                                                                                                                                                                                                                                           | IMAGER                           | F1000W | FASTR1          | 7                    | 3                        | 1               | Dither 1 | 4                       | 12                 | 255.304             |                              |
|                   | 3                                                                                            | LONG(C)                                                                                                                                                                                                                                                                                   | MRSLONG                          |        | FASTR1          | 7                    | 3                        | 1               | Dither 1 | 4                       | 12                 | 255.304             | 305707                       |
|                   | 3                                                                                            | LONG(C)                                                                                                                                                                                                                                                                                   | MRSSHORT                         |        | FASTR1          | 7                    | 3                        | 1               | Dither 1 | 4                       | 12                 | 255.304             | 305707                       |

|                      |                                                 |
|----------------------|-------------------------------------------------|
| Special Requirements | Sequence Observations 12, 13, Non-interruptible |
|----------------------|-------------------------------------------------|



# Proposal 12316 - Observation 13 - Probing the UV to MIR spectra of local Little Red Dots

|                   |                                                                                              |                  |                                                                                                                                                                                                                                                                                           |        |                 |                      |                  |                          |                 |               |                         |                     |                              |
|-------------------|----------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|----------------------|------------------|--------------------------|-----------------|---------------|-------------------------|---------------------|------------------------------|
| Observation       | Proposal 12316, Observation 13: Tol2240 MIRI MRS BKGD                                        |                  |                                                                                                                                                                                                                                                                                           |        |                 |                      |                  |                          |                 |               |                         |                     | Tue May 26 16:00:29 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                   |                  |                                                                                                                                                                                                                                                                                           |        |                 |                      |                  |                          |                 |               |                         |                     |                              |
|                   | Observing Template: MIRI Medium Resolution Spectroscopy                                      |                  |                                                                                                                                                                                                                                                                                           |        |                 |                      |                  |                          |                 |               |                         |                     |                              |
| Diagnostics       | (Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                  |                                                                                                                                                                                                                                                                                           |        |                 |                      |                  |                          |                 |               |                         |                     |                              |
| Fixed Targets     | #                                                                                            | Name             | Target Coordinates                                                                                                                                                                                                                                                                        |        |                 |                      |                  | Targ. Coord. Corrections |                 |               | Miscellaneous           |                     |                              |
|                   | (6)                                                                                          | TOL2240-384      | RA: 22 43 32.9629 (340.8873454d)                                                                                                                                                                                                                                                          |        |                 |                      |                  | Epoch of Position: 2016  |                 |               |                         |                     |                              |
|                   |                                                                                              |                  | Dec: -38 11 20.74 (-38.18909d)                                                                                                                                                                                                                                                            |        |                 |                      |                  |                          |                 |               |                         |                     |                              |
|                   |                                                                                              |                  | Equinox: J2000                                                                                                                                                                                                                                                                            |        |                 |                      |                  |                          |                 |               |                         |                     |                              |
|                   |                                                                                              |                  | Comments: Extent is 1"x2.5". Pointing center shifted NE by ~0.3" relative to peak, so IFU will cover point source and NE extension (based on LS DR10 imaging)<br>Category=Galaxy<br>Description=[Active galactic nuclei, Blue compact dwarf galaxies, Starburst galaxies]<br>Extended=YES |        |                 |                      |                  |                          |                 |               |                         |                     |                              |
| Acquisition       | #                                                                                            | Target           |                                                                                                                                                                                                                                                                                           |        |                 |                      |                  |                          |                 |               |                         |                     |                              |
|                   | 1                                                                                            | NONE             |                                                                                                                                                                                                                                                                                           |        |                 |                      |                  |                          |                 |               |                         |                     |                              |
| Template          | AcqFilter                                                                                    | Primary Channel  |                                                                                                                                                                                                                                                                                           |        |                 | Simultaneous Imaging |                  |                          | Imager Subarray |               | Grating Wheel Direction |                     |                              |
|                   |                                                                                              | All MRS          |                                                                                                                                                                                                                                                                                           |        |                 | YES                  |                  |                          | FULL            |               | Allow Auto Reorder      |                     |                              |
| Dithers           | #                                                                                            | Dither Type      |                                                                                                                                                                                                                                                                                           |        |                 |                      | Optimized For    |                          |                 | Direction     |                         |                     |                              |
|                   | 1                                                                                            | 4-Point          |                                                                                                                                                                                                                                                                                           |        |                 |                      | BACKGROUND       |                          |                 | NEGATIVE      |                         |                     |                              |
| Spectral Elements | #                                                                                            | Wavelength Range | Detector                                                                                                                                                                                                                                                                                  | Filter | Readout Pattern | Groups/Int           | Integrations/Exp | Exposures/Dith           | Dither          | Total Dithers | Total Integrations      | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                            |                  | IMAGER                                                                                                                                                                                                                                                                                    | F560W  | FASTR1          | 20                   | 3                | 1                        | Dither 1        | 4             | 12                      | 688.21              |                              |
|                   | 1                                                                                            | SHORT(A)         | MRSLONG                                                                                                                                                                                                                                                                                   |        | FASTR1          | 20                   | 3                | 1                        | Dither 1        | 4             | 12                      | 688.21              | 305707                       |
|                   | 1                                                                                            | SHORT(A)         | MRSSHORT                                                                                                                                                                                                                                                                                  |        | FASTR1          | 20                   | 3                | 1                        | Dither 1        | 4             | 12                      | 688.21              | 305707                       |
|                   | 2                                                                                            |                  | IMAGER                                                                                                                                                                                                                                                                                    | F770W  | FASTR1          | 13                   | 3                | 1                        | Dither 1        | 4             | 12                      | 455.107             |                              |
|                   | 2                                                                                            | MEDIUM(B)        | MRSLONG                                                                                                                                                                                                                                                                                   |        | FASTR1          | 13                   | 3                | 1                        | Dither 1        | 4             | 12                      | 455.107             | 305707                       |
|                   | 2                                                                                            | MEDIUM(B)        | MRSSHORT                                                                                                                                                                                                                                                                                  |        | FASTR1          | 13                   | 3                | 1                        | Dither 1        | 4             | 12                      | 455.107             | 305707                       |
|                   | 3                                                                                            |                  | IMAGER                                                                                                                                                                                                                                                                                    | F1000W | FASTR1          | 7                    | 3                | 1                        | Dither 1        | 4             | 12                      | 255.304             |                              |
|                   | 3                                                                                            | LONG(C)          | MRSLONG                                                                                                                                                                                                                                                                                   |        | FASTR1          | 7                    | 3                | 1                        | Dither 1        | 4             | 12                      | 255.304             | 305707                       |
|                   | 3                                                                                            | LONG(C)          | MRSSHORT                                                                                                                                                                                                                                                                                  |        | FASTR1          | 7                    | 3                | 1                        | Dither 1        | 4             | 12                      | 255.304             | 305707                       |

Proposal 12316 - Observation 13 - Probing the UV to MIR spectra of local Little Red Dots

|                      |                                                                                                                                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | Aperture PA Range 0.0 to 300.0 Degrees (V3 0.0 to 300.0)<br>Aperture PA Range 310.0 to 359.999 Degrees (V3 310.0 to 359.999)<br>Offset -40.0 arcsec, -45.0 arcsec<br><br>Sequence Observations 12, 13, Non-interruptible |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|